

NeuroSense: Leakage-Aware, Explainable EEG Decoding

An honest look at when EEG machine learning generalizes, and when it only appears to

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Abstract

Machine learning on EEG is easy to evaluate incorrectly. Because EEG is a time series with strong short-range correlation, randomly splitting a single continuous recording into training and test sets lets a model score highly by recognizing near-duplicate neighbors rather than by learning a generalizable mapping. This report studies that failure across two datasets and two paradigms, under a single prespecified model and an evaluation protocol whose unit of inference is the subject or the label block, not the epoch. On the UCI EEG Eye State dataset, a naive random split gives a balanced accuracy of 0.533, a chronological 70/30 holdout gives 0.416 (well below the 0.767 majority baseline), an expanding-window evaluation across label blocks gives 0.380 (95% CI 0.230 to 0.545), and a pre-specified leave-one-merged-block-out (super-block size 4 native blocks, declared before evaluation) gives balanced accuracy 0.456 (95% CI 0.270 to 0.605) and AUC 0.533 (95% CI 0.396 to 0.629); a `DummyClassifier(strategy="prior")` on the same merged splits scores balanced accuracy 0.400 (95% CI 0.200 to 0.500) and AUC 0.500 (95% CI 0.500 to 0.500), so the primary model is within a fold of the dummy. The retained leave-one-native-block-out diagnostic gives pooled AUC 0.437 for the real model and 0.000 for the same dummy on the same splits, which shows that the below-chance pooled AUC is a splitting artefact rather than electrode drift. Every leakage-safe Phase 1 protocol reports its own baselines, and the temporal gap between train and test is chosen from a measured feature-autocorrelation diagnostic. On the PhysioNet Motor Movement/Imagery dataset, within-subject shuffled-trial CV reaches a mean per-subject balanced accuracy of 0.607 (95% CI 0.522 to 0.700) and AUC of 0.632 (95% CI 0.534 to 0.735), with 8 of 10 subjects above 0.5 on both (a descriptive count, not a significance claim). A newly added within-subject leave-one-run-out evaluation across the three imagery runs gives balanced accuracy 0.588 (CI 0.529 to 0.648) and AUC 0.657 (CI 0.598 to 0.718) across 30 folds; it tests transfer across recording runs of the same session but does not establish transfer to a new session. Cross-subject leave-one-subject-out sits at chance (balanced accuracy 0.474, CI 0.432 to 0.516; AUC 0.511, CI 0.447 to 0.583), and per-fold labels, predictions, and probabilities are persisted in the results JSON so that a reader can rescore without re-fitting. Feature attribution recovers the expected C3/C4 sensorimotor lateralization in Phase 2. In Phase 1 it recovers nothing interpretable, because the model it describes performs near a same-split dummy; an earlier version of this work read those attributions as evidence of ocular artifact, and that reading is withdrawn here. Across both datasets, evaluation design determines apparent performance, and an explanation inherits the credibility of the evaluation beneath it.

1 Introduction

EEG underpins brain-computer interfaces, attention and fatigue monitoring, and a broad range of neurotechnology. Applying machine learning to EEG, however, requires care that is often skipped in introductory work: the signal is time-dependent, and samples that are close in time are highly

correlated. When such samples are split randomly into training and test sets, a model can appear highly accurate while mostly recognizing repeated temporal patterns it has already seen, rather than learning structure that transfers to new data.

This report treats that pitfall as the object of study rather than an afterthought. It asks two questions. First, do the high accuracies commonly reported on a popular EEG benchmark survive leakage-aware evaluation? Second, on a dataset built for generalization, how far does honest decoding actually go, and do the model’s explanations agree with known neurophysiology? The work is deliberately framed around evaluation validity and interpretability rather than headline accuracy, because on EEG a high number is only as trustworthy as the protocol that produced it.

A note on provenance. An earlier version of this report presented both phases with separate feature pipelines, different models per phase, and pooled epoch-level metrics. The analysis was subsequently rebuilt on a shared core with one band definition, preprocessing fitted inside each fold, and subject-level inference. Three claims from that earlier version did not survive the rebuild and are marked as withdrawn where they appear. A follow-up audit in September 2026 added a fold-safe regression guard, a feature-autocorrelation diagnostic that drives the choice of temporal gap, an expanding-window evaluation across label blocks, and per-split baselines. A second audit round replaced the single-class leave-one-block-out primary group protocol with a pre-specified leave-one-merged-block-out, paired every group protocol with a same-splits `DummyClassifier(strategy="prior")` diagnostic, added within-subject leave-one-run-out to Phase 2, persisted per-fold labels and predictions in the Phase 2 JSON, and tightened reporting language throughout: “at chance” is not used without a protocol-specific baseline, “N of 10 above 0.5” is labelled as a descriptive count rather than a significance claim, and the standard deviation across subjects is reported separately from the bootstrap CI of the mean. The audits changed no primary model and no substantive conclusion. Pre-audit results and figures are preserved verbatim under `reports/results/archive/` and `reports/figures/archive/`, and the change list is recorded in `reports/AUDIT_LOG.md`.

2 Data and Methods

Phase 1: UCI EEG Eye State. A single continuous 117-second recording from one subject, 14 channels sampled at 128 Hz, labeled for eyes open versus closed. The label runs in only 24 contiguous blocks with a median duration of about 3.9 seconds, of which 19 contain at least one complete window. Several channels contain single-sample electrode pops reaching roughly 700,000 against a baseline near 4,000; these are clipped per channel, using thresholds fitted on training data only. The signal is segmented into one-second non-overlapping windows, yielding 100 windows and 56 band-power features. Every Phase 1 figure below therefore rests on 100 observations from a single person.

Phase 2: PhysioNet Motor Movement/Imagery. Imagined left versus right fist movement across 10 subjects at 160 Hz, 383 trials in total drawn from runs R04, R08, and R12 (the pre-audit cache had 437 trials; the difference is that an older MNE version retained more boundary-adjacent epochs after concatenation, and the round-2 re-extraction with run IDs preserved uses MNE 1.8’s stricter default). Features are mu (8 to 13 Hz) and beta (13 to 30 Hz) band power over a 13-channel sensorimotor strip, giving 26 features per trial. The imagery-run identifier (R04, R08, or R12) is preserved per trial, so a leave-one-run-out evaluation is possible within subject.

Features. For both phases, band power is computed per channel and window using Welch’s method and integrated over each band. The delta, theta, alpha, and beta bands are used in Phase 1; the motor-relevant mu and beta bands in Phase 2. The lowest band edge is 1 Hz rather than 0.5 Hz,

because a one-second Welch segment resolves frequency only to 1 Hz. The gamma band (30 to 45 Hz) is excluded, since on consumer-grade hardware that range is dominated by muscle activity rather than cortical signal.

Model. A single model is prespecified for both phases and all headline results: logistic regression with $C = 1.0$, balanced class weights, and a fixed random seed. Support vector machines and random forests appear only as secondary comparisons. Fixing one model in advance avoids the failure mode, present in an earlier version of this work, of assembling a headline result from the strongest metric of whichever model produced it.

Evaluation. All preprocessing, including artifact clipping, band-power extraction, and feature scaling, is fitted inside the cross-validation fold. A regression test in the test suite mutates several held-out rows to extreme values, refits the Phase 1 pipeline on the same training rows, and asserts that the fitted clipper thresholds, the fitted scaler statistics, and the transformed training features are bit-identical to a baseline without the mutation. For Phase 1 the grouping unit is the contiguous label block, since the dataset contains one subject; for Phase 2 it is the subject. In both cases one model is fitted per held-out group and that group’s metrics are stored as a single row, so the headline figure is a mean over groups with a 95% percentile bootstrap interval that resamples groups. Pooled epoch-level metrics are reported only as a descriptive appendix: pooling correlated epochs and treating them as independent both understates uncertainty and, for threshold-free metrics, distorts the estimate itself. Every leakage-safe split reports its own baselines (majority-class, stratified dummy, uniform dummy) computed on that split’s test set alone; no baseline is shared across protocols. A naive shuffled split is included in each phase purely as a contrast, to quantify the inflation that leakage produces.

Temporal gap. For Phase 1 the choice of temporal gap between train and test in the chronological and expanding-window protocols is derived from a measurement rather than picked by hand. Mean absolute autocorrelation of the fold-safe band-power features is computed at lags 1 through 5; the recommended gap is the smallest lag at which that value first drops below a fixed threshold of 0.30. On this recording every lag sits in $[0.07, 0.10]$, so the recommendation is 0 windows. This is a genuine result about the feature representation on 14-channel consumer EEG at 128 Hz with one-second Welch integration: adjacent windows are effectively uncorrelated, and the chronological split does not need a buffer to be leakage-safe under this representation. The measurement, the threshold, and the recommendation are all recorded in the results JSON so a reviewer can trace them.

3 Results

4 Phase 1: Eye-State Detection

Because the label runs in long blocks, neighboring samples are near-identical and share a label. A random split scatters these neighbors across training and test, so a model scores well by recognizing duplicates. Holding everything else fixed and changing only how the split is drawn produces the performance gap shown in Figure 1.

The leakage-safe protocols agree that nothing survives. The chronological holdout gives a balanced accuracy of 0.416, which beats a stratified dummy (0.366) by a small margin but sits well below the 0.767 majority-class baseline. Expanding-window across 13 forward-walked label blocks gives 0.380 (0.230 to 0.545), near a stratified dummy. The pre-specified leave-one-merged-block-out primary protocol gives balanced accuracy 0.456 (0.270 to 0.605) and AUC 0.533 (0.396 to 0.629), while

Phase	Protocol	Bal. accuracy	Macro F1	ROC-AUC
1	Naive random window split	0.533	0.533	0.566
1	Chronological 70/30 holdout (gap = 0)	0.416	0.330	0.491
1	Expanding-window (13 folds, gap = 0)	0.380 [0.230, 0.545]	0.283 [0.165, 0.432]	undefined
1	Leave-one-merged-block-out (5 folds, primary)	0.456 [0.270, 0.605]	0.433 [0.255, 0.582]	0.533 [0.396, 0.629]
1	Same splits, DummyClassifier(prior) diagnostic	0.400 [0.200, 0.500]	0.269 [0.123, 0.374]	0.500 [0.500, 0.500]
1	Leave-one-native-block-out (19 folds, diagnostic)	0.482 [0.334, 0.630]	0.369 [0.239, 0.515]	undefined
2	Naive random trial split	0.504	—	0.521
2	Within-subject shuffled trial CV	0.607 [0.522, 0.700]	0.605 [0.518, 0.698]	0.632 [0.534, 0.735]
2	Within-subject leave-one-run-out (30 folds)	0.588 [0.529, 0.648]	0.556 [0.490, 0.622]	0.657 [0.598, 0.718]
2	Cross-subject (LOSO)	0.474 [0.432, 0.516]	0.427 [0.380, 0.474]	0.511 [0.447, 0.583]

Table 1: All results under the prespecified model. Brackets give 95% percentile bootstrap intervals re-sampling blocks or super-blocks (Phase 1) or subjects (Phase 2). Per-split baselines on the Phase 1 chronological test segment: majority-class 0.767, stratified dummy 0.366, uniform dummy 0.339. On the naive random Phase 1 split the majority baseline is 0.550. Native leave-one-block-out and expanding-window folds are single-class by construction, so their majority-class is trivially 1.0 and the meaningful reference is chance (0.5). Leave-one-merged-block-out uses 4 native blocks per super-block, a rule declared in `src/phase1_eyestate.py` before evaluation; 4 of 5 super-blocks contain both classes so per-fold AUC is defined. Within-subject leave-one-run-out on Phase 2 is added by the 2026-09 audit; see `reports/results/phase2_results.json` for its per-fold rows.

a `DummyClassifier(strategy="prior")` on the same splits gives balanced accuracy 0.400 (0.200 to 0.500) and AUC 0.500 (0.500 to 0.500); the two intervals overlap heavily and the point estimates differ by less than one fold, so this is not evidence of decoding. Leave-one-native-block-out is retained only as a diagnostic (single-class folds by construction, AUC undefined per fold). The naive random window split reaches 0.533, matched by its stratified dummy at 0.552 on the same test rows, which is the tell that the split rather than the model is doing the work.

The 2026-09 audit added a regression test that fabricates a held-out epoch with extreme values, refits the Phase 1 pipeline on the training epochs, and requires the fitted clipper thresholds, the fitted scaler statistics, and the transformed training features to be bit-identical to a baseline without the mutation. A negative control confirms that this guard is not vacuous: a globally fitted clipper on the mutated data does shift its thresholds. Every Phase 1 fold row also carries a provenance block recording its test-window start-sample indices, block ids, and a hash of the training indices, so a fold in the JSON can be re-located in the raw recording without rerunning the code.

One structural caveat deserves emphasis. Every contiguous native label block is single-class by construction, so a held-out native block (in expanding-window or leave-one-native-block-out) contains only eyes-open or only eyes-closed windows. ROC-AUC cannot be computed on a single-class test set, and balanced accuracy degenerates into the recall of whichever class the block contains. An earlier version of this report gave a leave-one-block-out AUC range of 0.28 to 0.37 and read below-chance AUC as evidence that the features encode recording segment rather than eye state. That figure can only have come from pooling predictions across separately-trained folds, and the claim is withdrawn: the quantity it rests on is a known splitting artefact, not electrode drift. To make that concrete, the 2026-09 audit runs `DummyClassifier(strategy="prior")` on the same native splits: it scores pooled AUC 0.000 while learning nothing, exactly the below-chance pattern that would have been read as “drift”. The primary group protocol is now leave-one-merged-block-out with a super-block size of 4 native blocks, declared in `src/phase1_eyestate.py` before evaluation; 4 of 5 super-blocks contain both classes, so per-fold AUC is defined and the same-splits dummy comparison remains as a sanity check. Native LOBO is retained only as a labelled diagnostic.

Phase 1 | UCI EEG Eye State: apparent skill is an artifact of the split (100 windows, 1 subject)
feature autocorrelation lag1=0.09, lag2=0.09, lag3=0.10, lag4=0.07, lag5=0.08 → gap = 0 windows (already_below)

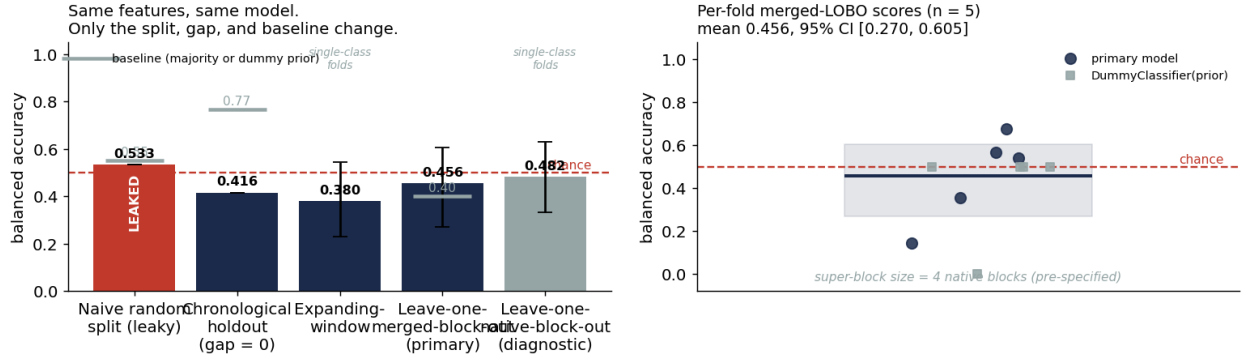


Figure 1: Left: balanced accuracy under five leakage-aware splits of the same features and model, with the chance level marked in red. The majority-class baseline is shown for the two splits where it is informative (naive at 0.55, chronological at 0.77); for leave-one-merged-block-out the DummyClassifier(prior) score on the same splits is shown (0.40); for expanding-window and native leave-one-block-out the caption annotates “single-class folds” because the majority-class baseline is trivially 1.0. The chosen temporal gap of 0 windows is derived from the feature-autocorrelation diagnostic shown in the header. Right: the per-fold merged-LOBO scores that the mean averages over, with both the primary model and the DummyClassifier(prior) plotted so the reader can see the two overlap.

4.1 Explainability, and its limits

A pre-modeling check found no clean Berger effect in this recording: occipital alpha does not rise on eye closure. An earlier version went further and reported that the most class-separating features were frontal, reading this as ocular and blink artifact captured by frontal electrodes. That claim is withdrawn. Under the prespecified pipeline the top ten features are six frontal and four posterior or temporal, and the single strongest is temporal alpha at T7.

More fundamentally, the model these attributions describe scores 0.416 balanced accuracy, well below the 0.767 majority-class baseline and only marginally above a stratified dummy. Permutation importance computed on a model that does not clear its own baselines is not evidence about physiology, and it is not evidence about artifact either. The honest conclusion is that Phase 1 yields no trustworthy explanation at all. That is itself the finding: an explanation inherits the credibility of the evaluation beneath it, and here there is none to inherit. What remains supported is the negative result, that a single continuous recording from one person cannot support generalizable eye-state decoding, and that accuracies reported on this dataset under random sample-level splits do not survive leakage-aware evaluation.

5 Phase 2: Motor Imagery

The same band-power approach is applied to data with many independent trials and multiple subjects. Two honest evaluations tell a two-part story (Figure 2). Within a subject, shuffled-trial cross-validation gives a mean per-subject balanced accuracy of 0.607 with a bootstrap CI of 0.522 to 0.700 and 8 of 10 subjects above 0.5 (a descriptive count, not a significance claim); the mean AUC is 0.632 with a CI of 0.534 to 0.735, and 8 of 10 subjects above 0.5. Both CIs narrowly exclude

0.5. The 2026-09 audit added a within-subject leave-one-run-out (LORO) evaluation, which trains on two of a subject’s three imagery runs (R04, R08, R12) and tests on the third. Across all 30 (subject, run) folds LORO gives balanced accuracy 0.588 (CI 0.529 to 0.648, 19 of 30 folds above 0.5) and AUC 0.657 (CI 0.598 to 0.718, 25 of 30 folds above 0.5). LORO is close to the shuffled-trial CV, which is reassuring: the within-subject signal is consistent across recording runs of the same session. LORO still does not establish transfer to a new recording session, because all three runs share the electrode montage, cap placement, and skin condition of a single day. Per-fold LORO rows and per-fold labels, predictions, and probabilities are stored in the results JSON.

Across subjects, leave-one-subject-out gives balanced accuracy 0.474 (CI 0.432 to 0.516, standard deviation across subjects 0.072) and AUC 0.511 (CI 0.447 to 0.583, std 0.115), both at chance. Standard deviation across subjects is the spread between people; the bootstrap CI is the uncertainty of that mean, computed by resampling subjects (not trials). Individual subjects range from 0.353 to 0.600, a spread that a single pooled figure conceals entirely. Every held-out subject’s y_{true} , y_{pred} , and y_{proba} are persisted in the JSON so the reader can rescore or diagnose without re-fitting.

An earlier version reported cross-subject AUC as 0.48 and described it as below chance. That figure pooled every held-out subject’s predicted probabilities into one array before scoring. Because subjects’ decision scores sit on different scales, pooling manufactures apparent below-chance performance; the per-subject mean is 0.511. The pooled scores are retained only as a labelled descriptive appendix.

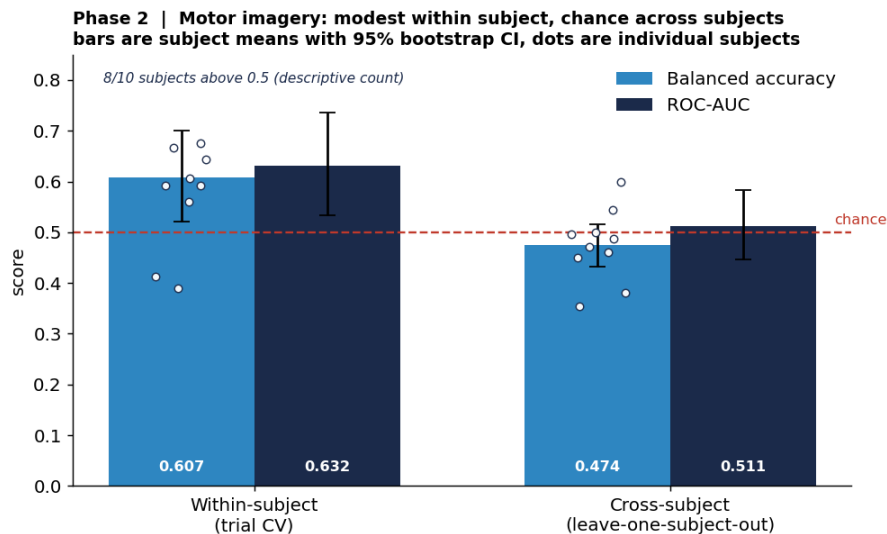


Figure 2: Within-subject decoding is modestly above chance on AUC; cross-subject transfer sits at chance. Bars are subject means with 95% bootstrap intervals; dots are the ten individual subjects.

The interpretability check here behaves very differently from Phase 1. The within-subject model keys on the C3 versus C4 mu rhythm (Figure 3, right), the lateralized sensorimotor pattern expected from contralateral desynchronization during motor imagery. Here the attribution corroborates known physiology on a model that genuinely decodes.

Explainability is only as trustworthy as the evaluation beneath it: uninterpretable where the model fails (left), corroborating known physiology where it works (right)

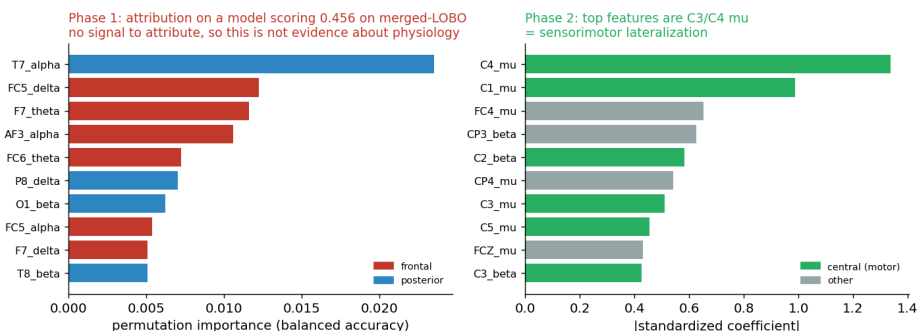


Figure 3: Explainability is only as trustworthy as the evaluation beneath it. Right: Phase 2 top features sit on the central motor strip, consistent with sensorimotor lateralization, on a model that decodes above chance. Left: Phase 1 attributions on a model whose merged-LOBO score (0.456) overlaps the same-splits dummy prior, and therefore not readable as a finding in either direction.

6 Discussion

Across both datasets, naive evaluation overstates performance, and leakage-aware evaluation reveals that EEG band-power features capture structure other than the intended target: time and session structure in Phase 1, subject identity in Phase 2. Honest within-subject motor-imagery decoding is achievable but modest, and where it works its explanations align with sensorimotor physiology.

One caveat cuts against a tempting generalization. The naive random split inflates dramatically in Phase 1 but not in Phase 2, where it reaches 0.504 against a cross-subject 0.474. Phase 2 uses discrete trials, and random splitting of discrete trials leaks far less than random splitting of continuous windows. How much a naive split inflates depends on how much temporal structure the epochs share, so the dramatic result belongs to Phase 1 and should not be stated as a general law of EEG machine learning.

Taken together, the two phases make a single point: model explanations are usable as evidence only when the evaluation beneath them is leakage-free. The corollary, learned the hard way in the course of this work, is that where the evaluation shows a model failing, its attributions should be reported as uninterpretable rather than mined for a plausible story. An attribution plot on a broken model is a confident description of nothing.

7 Limitations

Phase 1 is a single subject and single session recorded on consumer-grade hardware, so its results are not subject-generalizable and its high-frequency content is unreliable. Windowing produces only 100 windows, which is why every Phase 1 interval is wide, and its single-class label blocks make leave-one-block-out only partially interpretable. Phase 2 uses 10 of the 109 available subjects, so its intervals are wider than necessary. The two phases also differ in hardware, channel count, and sampling rate, so any comparison between them confounds paradigm with recording setup; the contrast between them is offered as a hypothesis rather than a controlled result.

8 Future Work

The natural next steps are to scale Phase 2 to all 109 subjects to tighten the estimates, to add common spatial pattern (CSP) features that are standard for motor imagery, and to attempt subject-adaptive transfer, both unsupervised alignment and small-sample calibration, to lift cross-subject decoding above chance. A third phase on mental arithmetic is planned, with its hypotheses, ablations, and primary endpoint fixed in advance in a committed preregistration. Deep models are deferred until these leakage-safe baselines are established.

References

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